

F). Now the equations have a coefficient of x^2 different than 1.

E.g. $2x^2 - 14x + 9 = 0$ Divide throughout to get rid of the 2 in front of the x^2 .
 $x^2 - 7x + 4.5 = 0$ Now carry on and solve as before.

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|--------------------------|--------------------------|--------------------------|
| 1). $4c^2 - 8c + 1 = 0$ | 2). $2x^2 - 14x + 9 = 0$ | 3). $2c^2 - 3c - 9 = 0$ |
| 4). $5e^2 + 15e + 1 = 0$ | 5). $3m^2 - 5m + 2 = 0$ | 6). $2d^2 + 10d + 5 = 0$ |
| 7). $3k^2 + 2 = 9k$ | 8). $5f^2 - 20f = 28$ | 9). $3a^2 - 12a = 2$ |
| 10). $4h^2 = 8h + 3$ | 11). $2b^2 = 8b + 11$ | 12). $3m^2 = 6m + 2$ |

Finding a Formula by Completing the Square

If we use the techniques we have been using for completing the square, we can find a formula to solve quadratics with.

We have the quadratic $ax^2 + bx + c = 0$, where a , b and c are numbers.

First we get rid of the a in front of x^2 ($\div a$) $x^2 + \frac{bx}{a} + \frac{c}{a} = 0$

Then get rid of the 'number' term ($-\frac{c}{a}$) $x^2 + \frac{bx}{a} = -\frac{c}{a}$

Complete the square ($+\left(\frac{b}{2a}\right)^2$) $x^2 + bx + \left(\frac{b}{2a}\right)^2 = -\frac{c}{a} + \left(\frac{b}{2a}\right)^2$

Work out the last bracket $x^2 + bx + \left(\frac{b}{2a}\right)^2 = -\frac{c}{a} + \frac{b^2}{4a^2}$

Factorise the left-hand side $\left(x + \frac{b}{2a}\right)^2 = -\frac{c}{a} + \frac{b^2}{4a^2}$

Tidy up the right-hand side. $\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$

Square root both sides $x + \frac{b}{2a} = \pm \sqrt{\frac{b^2 - 4ac}{4a^2}}$

Get x on its own $x = -\frac{b}{2a} \pm \sqrt{\frac{b^2 - 4ac}{4a^2}}$

So for a quadratic

$$ax^2 + bx + c = 0 \quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

where a , b and c are numbers.

